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Div-F

ROll no- 12

Aim - Program bassed on Functions

1. Lambda function (sum of two numbers)

```
def calc(a, b):
    return a+b, a-b

sum, diff = calc(10, 5)
print(sum, diff)
```

```
15 5
```

2. Function product (any number of arguments)

```
def product(*args):
    result = 1
    for num in args:
        result *= num
    return result

print(product(2, 3, 4))
```

```
24
```

3. Function print_info (keyword arguments)

```
def print_info(**kwargs):
    for key, value in kwargs.items():
        print(key, ":", value)

print_info(name="Viren", age=18, city="Pune")
```

4. Compound Interest (using keyword arguments)

```
def compound_interest(**kwargs):
    P = kwargs.get("principal")
    R = kwargs.get("rate")
    T = kwargs.get("time")

    CI = P * (1 + R/100) ** T
    return CI

print(compound_interest(principal=1000, rate=5, time=2))
```

5. Stack Operations using *args

```
def stack_operations(*args):
    stack = []

    for op in args:
        if op[0] == "push":
            stack.append(op[1])
        elif op[0] == "pop":
            if stack:
                print("Popped:", stack.pop())
            else:
                print("Stack is empty")

    print("Final Stack:", stack)

stack_operations(("push", 10), ("push", 20), ("pop",), ("push", 30))
```

6. Age calculation using **kwargs

```
from datetime import date

def calculate_age(**kwargs):
    birth_date = date(kwargs["year"], kwargs["month"], kwargs["day"])
    today = date.today()

    age = today.year - birth_date.year

    if (today.month, today.day) < (birth_date.month, birth_date.day):
        age -= 1

    print("Age:", age)

calculate_age(year=2007, month=11, day=7)
```